

Jurisdiction–Aware Credit Decision Governance Monitor

A senior–management pitch for Revolut’s AI–assisted credit decision governance across the US and EU

Option C: Analytical Design + Quantitative Prototype

Shared Governance Schema ≠ Shared Sensitive–Data Pipeline

Core product thesis

Same platform logic for evidence, version control and escalation — different jurisdiction–bound data access, computations and human review triggers.

Scope discipline
The tool produces governance evidence and risk signals. It does not make credit decisions, issue legal opinions, certify EU AI Act conformity or convert SHAP values into final adverse–action notices.

Assignment artefacts used
Task 3 design report, jurisdiction config, regulatory mapping table, synthetic data, US/EU evidence outputs, sensitivity results and tool governance card.

What senior management needs to understand before approving a RegTech product boundary.

1 **Regulatory divergence**
US and EU do not ask for the same model report. They encode different political choices about explanation, bias testing and lifecycle governance.

2 **Product boundary**
The product deliberately avoids a unified sensitive–data pool. Governance is shared; data paths are not.

3 **Operational proof**
Synthetic data and rule configuration show how the tool emits different evidence packages under US and EU settings.

4 **Honest limits**
The tool supports evidence and escalation. It does not certify legality, make credit decisions or automatically repair models.

Management takeaway

Approve the tool as a governance monitor, not as a credit decision engine. Its value is in evidence routing, rule versioning and accountability under jurisdictional divergence.

Business problem: AI credit governance is not one global checklist

Revolut-like digital credit products scale faster than jurisdiction-specific evidence workflows.

Consumer impact

Credit decisions shape access to essential financial services. If the model fails, harm scales quickly and can remain hidden behind apparently neutral variables.

Compliance impact

Regulators expect evidence that the institution can explain decisions, monitor bias, maintain logs and route judgement to accountable owners.

Operating impact

Instant decisioning is valuable, but broad manual review can paralyse the business. Oversight must be risk-tiered, not indiscriminate.

Why the current memo/spreadsheet approach fails

Rule version	Data path	Metric logic	Evidence owner	Escalation
Executable, logged and reviewable	Enforced by contract, not prose	Jurisdiction-bound calculations	Named accountable owner	Risk-tiered workflow

Board-level question changes from “Did someone write a memo?” to “Which evidence package, rule version, owner and escalation level support this workflow?”

Regulatory divergence: different political choices, different tool requirements

The same credit decisioning problem does not produce the same evidence obligation.

Dimension	United States	European Union	Design implication
Primary question	Can specific and accurate reasons be provided for adverse action?	Can high-risk AI governance be evidenced before and after deployment?	Different computation modules, not report skins.
Sensitive data posture	US live path hard-drops audit-only protected fields.	Audit-only sensitive fields only in segregated, lawful testing path.	Shared schema, separate data paths.
Explanation posture	SHAP-style attribution is internal aid; approved reason codes are reviewed.	Explanation forms part of broader lifecycle governance.	Evidence packages differ by jurisdiction.
Human judgement	Low-confidence reasons and proxy-risk signals trigger review.	Governance gaps, bias tests and drift trigger review.	Same ladder, different triggers.

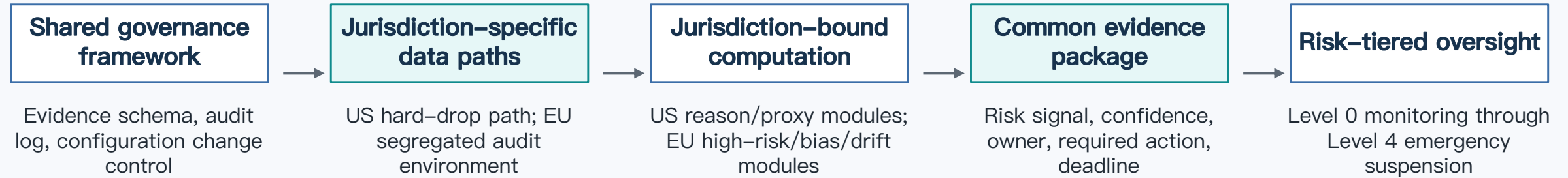
Do not flatten the regimes into one global compliance score.

The tool keeps management reporting comparable while preserving legal difference: common evidence fields; jurisdiction-bound data access and computations.

Product architecture: shared governance, jurisdiction-bound computation

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What is shared is the governance evidence system, not the sensitive-data pipeline.



Why this is not a decorative selector

The jurisdiction choice changes data contracts, allowed fields, computation modules, evidence outputs and review owners. It is not a front-end dropdown that swaps labels.

Why this is commercially usable

Human review is tiered. Minor drift creates a monitoring ticket; material governance gaps or data leakage trigger deployment hold or emergency suspension.

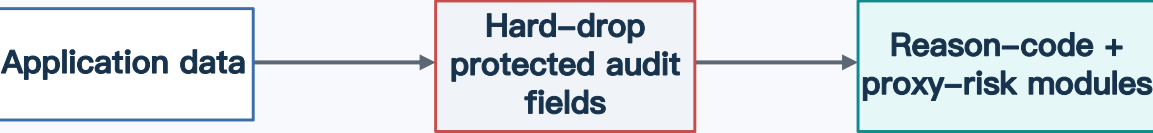
Unified evidence schema fields

jurisdiction • model_version • data_version • rule_version • risk_signal • evidence_generated • confidence_level • owner • action • deadline • escalation_level • audit_log_reference

Data architecture: the product boundary is a data boundary

US and EU modules share a platform but not a sensitive-data pool.

US hard-drop path



No age, gender_audit, age_band, protected_group_flag or audit_group in Task3_US_Processed_Data.csv.

EU segregated audit path



Audit-only fields enter bias testing only with lawful basis, access controls, logs, data minimisation and retention limits.

Data file	Rows	Protected/audit fields	Purpose
Task3_Synthetic_Credit_Data.csv	1,400	Contains synthetic audit fields	Master synthetic dataset
Task3_US_Processed_Data.csv	700	Hard-dropped	US live governance path
Task3_EU_Audit_Data.csv	700	Audit-only fields retained	EU controlled bias-testing path

Rule engine: legal–policy mapping becomes executable governance

Every computation has a rule owner, version, data–access posture and rollback path.

Rule	Module	Owner	Trigger
US-ADV-001	Reason-code mapping	US Compliance	Low confidence < 0.58
US-PROXY-002	Proxy-risk monitor	Fair Lending Counsel	Postcode/thin-file gaps
US-DATA-003	Data contract validator	Data Governance	Protected field appears
EU-HRAI-001	Pre-deployment gate	EU AI Governance	Completeness < 90%
EU-BIAS-002	Controlled audit bias	EU Model Risk	Material group gap
EU-DRIFT-003	Drift monitoring	Model Monitoring	PSI/material shift

Configuration controls

Each rule stores: rule_id, jurisdiction, legal_basis, data_access, computation_module, output_required, human_review_trigger, owner, version, effective_date, approval_status and rollback mechanism.

Why this matters

A rule change is not an invisible code edit. It becomes a governed event with owner, approvals, versioning and rollback. That directly addresses jurisdiction–configuration drift risk.

Design discipline

The legal mapping table is the backbone; jurisdiction_config is the executable layer. If the law/policy basis is uncertain, the rule is labelled as internal governance rather than invented statute.

Quantitative prototype: synthetic data proves the governance mechanics

The numbers are synthetic; the routing, evidence and escalation logic are the deliverable.

Applications	US path	EU path	US declines	Evidence rows	EU checklist
1,400	700	700	521	6	87.5%
Synthetic applications	Processed records	Audit records	Reason-code candidates	US/EU packages	Pre-deploy completeness

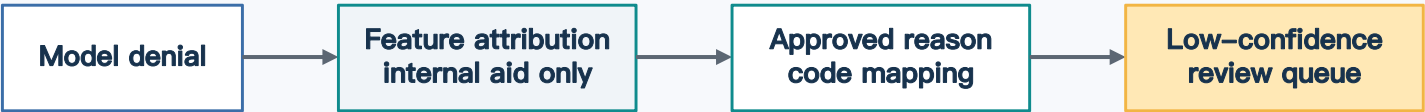
Shared evidence schema, different risk signals

Jurisdiction	Risk signal	Owner	Action
US	Adverse action reason-code quality	US Compliance Lead	Review low-confidence reason samples
US	Potential proxy discrimination signal	Fair Lending Counsel	Open privileged fair lending review
EU	High-risk AI evidence gate	EU AI Governance Lead	Do not approve launch until owner assigned
EU	Controlled audit bias testing	EU Model Risk Lead	Document remediation decision

US module deep dive: explanation quality + proxy-risk review

US path does not use direct protected attributes; it detects risk through reason quality and proxy signals.

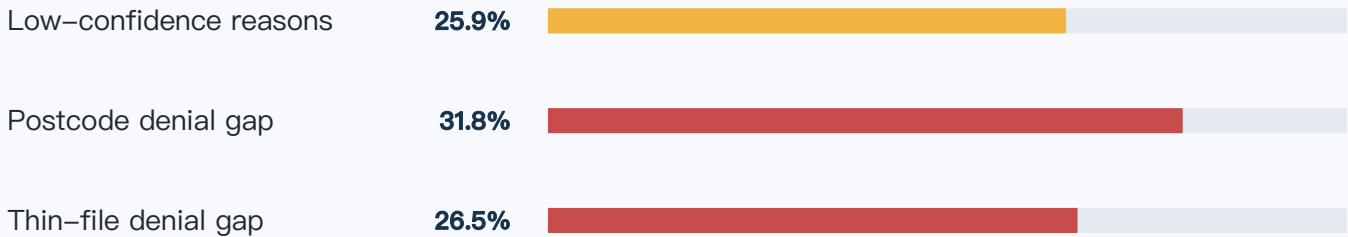
Reason-code workflow



Important boundary

Postcode risk is an internal fair-lending review signal, not a consumer-facing denial reason. SHAP-style attribution helps identify candidate factors but is not legally sufficient by itself.

Core US metrics



Evidence output

US evidence package assigns owner, deadline and escalation level: reason-code review to US Compliance; proxy-risk review to Fair Lending Counsel; hard-drop control to Data Governance.

Reason code	Count	Low-conf. rate
AA-CR-01	209	8.1%
AA-DTI-02	77	1.3%
AA-EMP-03	25	0.0%
AA-INC-06	26	0.0%

EU module deep dive: high-risk lifecycle governance

EU creditworthiness AI enters a high-risk governance workflow before and after deployment.

Pre-deployment gate

The tool does not “discover” high-risk status after the fact. Creditworthiness / credit score use is treated as high-risk and the gate checks evidence completeness before approval.

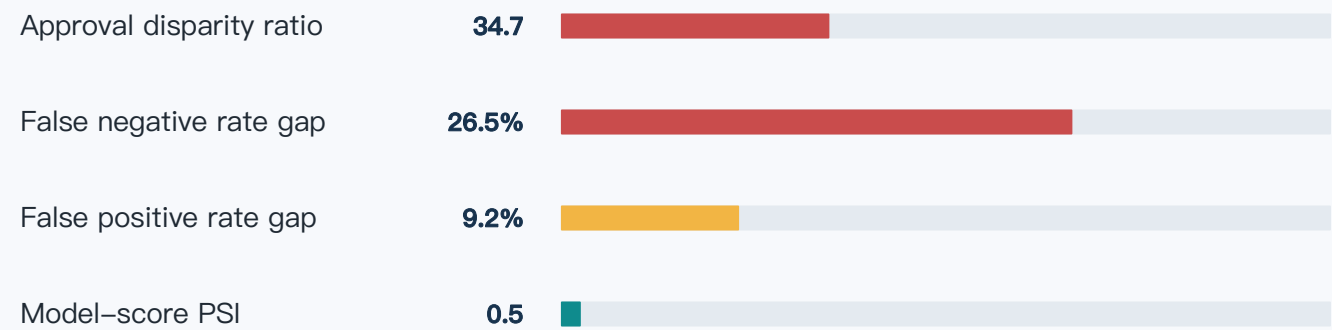
Controlled audit bias testing

Audit-only sensitive fields are processed only in a segregated environment with lawful basis, access control, logs, minimisation and retention controls.

Post-deployment monitoring

The tool tracks score drift, DTI shift and group-level performance changes to decide whether monitoring, sampled review or model governance review is needed.

EU metrics snapshot



Pre-deployment checklist

Evidence item	Complete
Intended purpose and high-risk classification recorded	Yes
Training data provenance documented	Yes
Data governance controls documented	Yes
Bias testing plan approved	Yes
Human oversight workflow approved	Yes

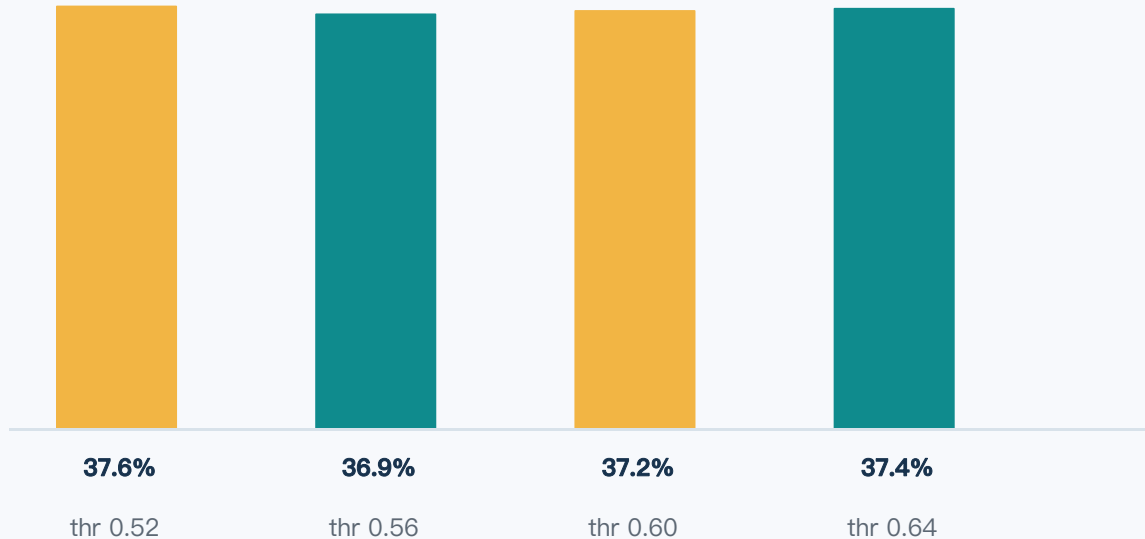
Completeness = 87.5%; missing owner blocks EU deployment approval in the prototype.

Sensitivity analysis: governance strength vs operating friction

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Option C requires showing how conclusions move when thresholds or data distribution shift.

Human review workload under threshold changes



Interpretation

A stricter model or lower alert threshold increases review workload. The tool responds with Level 1 monitoring or Level 2 sampled review before considering business-blocking action.

Business design choice

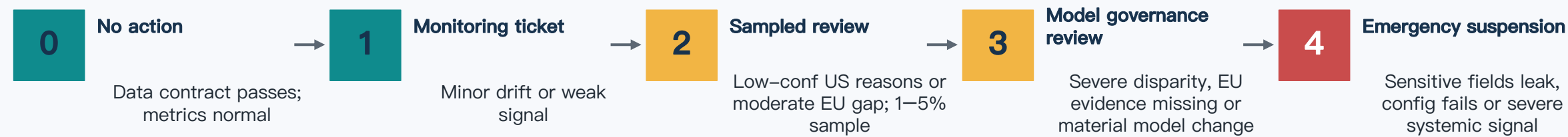
Governance cannot become an operational denial-of-service. The monitor reserves Level 3/4 interventions for governance evidence failure, sensitive-field leakage or severe systemic risk.

Sensitivity scenarios included

Approval threshold changes • alert threshold changes • protected-group DTI shift • human-review workload estimation

Human oversight: judgement is routed, not outsourced

The tool escalates by risk tier to protect both governance quality and instant decisioning.



Trigger type	Typical owner	Why human judgement is needed
Low-confidence reason code	US Compliance Lead	Consumer-facing reason requires more than post-hoc attribution.
Proxy-risk signal	Fair Lending Counsel	Proxy patterns need privileged legal/model-risk interpretation.
EU evidence gap	EU AI Governance Lead	Deployment approval depends on governance completeness.
Sensitive-field leakage	Data Governance + CCO	This is a data-path failure, not just a metric failure.

Failure modes: what can go wrong, who is harmed, and what the tool does next

The strongest work names failure modes specific to RegTech, not just generic model risk.

Failure mode	Who is harmed	Detection	Mitigation	Residual risk
Jurisdiction config drift	Applicants + compliance owners	Version mismatch or field in wrong path	Four-eyes approval, audit log, rollback	A bad policy can still be approved.
US reason-code failure	Declined applicants	Low-confidence or REVIEW-PROXY reason	Sampled review + reason taxonomy	Post-hoc explanation is imperfect.
US proxy blind spot	Protected communities	Postcode/thin-file concentration	Privileged fair lending review	No direct labels can under-detect harm.
EU audit data failure	EU applicants	Access logs + data contract tests	Segregated audit environment	Audit path increases privacy risk.
Human oversight overload	Applicants + operations	Workload sensitivity analysis	Risk-tiered review	Low thresholds can slow approvals.

Management message

The tool reduces model-governance risk but creates its own configuration, data-access and evidence-reliance risks. It therefore needs governance as a product, not just validation as a dashboard.

Strategic boundaries and roadmap

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The product is valuable because it is disciplined about what it refuses to automate.

No unified sensitive–data pool

Cost: weaker cross–jurisdiction comparability; higher engineering cost. Reason: avoid US contamination and cross–jurisdiction permission confusion.

No automatic final adverse action notice

Cost: human review remains necessary. Reason: SHAP–style attribution is not legally sufficient by itself.

No automatic model repair

Cost: detection–to–remediation time lag. Reason: automatic changes break governance and audit trails.

No legal conclusion / certification

Cost: legal and compliance staff remain involved. Reason: tool provides evidence and escalation, not legal judgement.

Roadmap

- 1 Connect to real Revolut or partner–bank credit data
- 2 Add legal–rule update workflow
- 3 Add immutable audit logging
- 4 Add issue tracker integration
- 5 Add production model drift dashboards
- 6 Add UK / Singapore jurisdiction packs

Recommendation

Approve as a governance monitor for evidence generation and escalation. Do not position it as a credit engine, legal–certification tool or automated bias–remediation system.

Submission package: what this deck is tied to

The detailed deck is designed to be edited and aligned with the Task 3 artefacts in the zip.

Artefact	Role in senior–management story
Task3_Tool_Design_Report.pdf	Full design rationale, failure modes, boundaries and quantitative interpretation.
Task3_Jurisdiction_Config.json	Executable rule layer with owner, version, legal basis and rollback.
Task3_Regulatory_Mapping_Table.pdf/csv	Law/policy–to–tool mapping that prevents decorative jurisdiction logic.
Task3_Quantitative_Analysis_Notebook.ipynb/html	Reproducible synthetic data and evidence–output generation.
Task3_Output_US/EU_Evidence.csv	Shared evidence schema with jurisdiction–bound risk signals and owners.
Task3_Sensitivity_Results.csv	Threshold and workload analysis for governance vs business friction.

Optional enhancement for figures folder

Generate chart images from the notebook and place them under task3/figures/. The deck itself uses editable PowerPoint charts/shapes, but the submission package should include reusable static figures for readers.

File naming discipline

Final editable deck should be named Task3_Senior_Management_Presentation_Detailed.pptx. If exported to PDF, use the same Task3_ prefix.